
Sharper Frames, Worse Rooms: Viewpoint Coverage Dominates Image Quality in Feed-Forward 3D Reconstruction

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Abstract

Grounded spatial reasoning about indoor scenes rests on a geometric reconstruction that the multimodal system does not itself produce. Feed-forward multi-view reconstruction models operate under a fixed view budget, making frame selection a critical upstream decision. We show that under this setting the conventional preference for sharp frames can be actively harmful. Across nine handheld room scans, selecting the sharpest 16 of 200 candidate frames is the worst strategy we test, reaching 48% room dimension error where uniform temporal sampling reaches 3%. We trace this failure to a coupling between image quality and camera motion: sharp frames tend to occur during slow motion and therefore concentrate the selection in a narrow angular region, leaving 192° of the room unobserved in the case we illustrate and over 300° in the worst cells. A 2×2 factorial experiment that independently manipulates viewpoint spread and per-frame sharpness shows that spread has roughly three times the main effect of quality, and that blurry but well spread selections outperform sharp but clustered selections in 8 of 9 captures. The advantage also diminishes as the frame budget increases and angular coverage saturates, consistent with this mechanism. Finally, we ask whether the failure is detectable at inference: the reconstruction model’s multi-view consistency score increases with reconstruction error, while a vision-language model given the same frames reports confidence unrelated to its own spatial estimation error. Under small view budgets, viewpoint coverage is a more important constraint than per-frame image quality, and neither uncertainty signal available at inference reveals this upstream failure.

1 Introduction

A phone scanning a room produces on the order of 1,800 frames. The reconstruction model consuming them accepts sixteen. Something has to choose, and that choice fixes the geometry every downstream task inherits.

The literature on this choice is mature. Classical keyframe selection for structure-from-motion maximises triangulation baseline, avoids degenerate configurations, and rejects motion blur [1, 8, 12]. Later work adds learned blur classifiers [5], blur metrics coupled to flight geometry [14], and appearance change signals [3]. Across all of it, per-frame image quality is something to maximise and blur is something to remove. We found no prior work that evaluates this trade-off under a fixed frame budget in this capture regime.

We ask that question in a setting where the answer turns out to matter. Our reconstruction backbone is MapAnything [4], a feed-forward transformer that ingests N views jointly and regresses metric geometry. It computes no correspondences, estimates no fundamental matrix, and runs no bundle

adjustment. These are substantially different assumptions from the pipelines the classical criteria were designed for, which motivates testing whether the criteria transfer. Note also that the configurations those criteria explicitly reject, namely rotation about the camera centre and coplanar structure, describe a handheld sweep of a room with flat walls.

Our hypothesis is that under a fixed frame budget and rotation dominant handheld capture, reconstruction error is governed more by viewpoint coverage than by per-frame image quality. The hypothesis was motivated by observing that sharpness ranked selection failed badly, so §4.1 and §4.2 are confirmatory rather than predictive; §4.3 tests a prediction made before running it, that the advantage of spread should shrink once any reasonable selection covers the scene.

The reliability angle is what makes this a deployment problem rather than a tuning problem. A pipeline that silently loses viewpoint coverage does not merely become less accurate. It produces geometry that is confidently wrong, and the signal a practitioner would reach for to detect that, the model’s own multi-view consistency score, moves in the wrong direction. Downstream multimodal reasoning about the scene inherits the corrupted geometry with no indication that anything failed.

The budget is not a matter of taste either. The deployed system we study serves a reconstruction graph exported at a fixed sixteen views, so selection is a hard cardinality constraint imposed by the architecture.

Our contributions:

- Sharpness ranked selection is the worst performing strategy we test, worse than random at $K=16$ and worse at covering the scene at every budget, and we identify the mechanism: sharp frames co-occur with slow camera motion, so ranking by sharpness ranks by angular redundancy.
- A 2×2 factorial isolates that mechanism, manipulating spread and quality independently. Both matter; spread matters about three times more.
- The effect has a predictable boundary. Sweeping K from 8 to 48, both the error gap and the catastrophic failure rate collapse where angular coverage saturates.
- Neither confidence signal available to a practitioner detects the failure. The model’s multi-view consistency score increases with error in every capture we test, and a VLM’s self-reported confidence carries no usable information about its own error.

2 Related Work

Triangulation-era criteria. Ahmed et al. [1] select keyframes using a correspondence ratio as a baseline proxy, Torr’s GRIC [12] to choose between homography and fundamental matrix models, and a point-to-epipolar-line cost, building on earlier correspondence-ratio and reprojection-error criteria [10, 11] and on hand-held camera modelling [8]. Their stated motivation is triangulation accuracy: uncertainty grows as inter-view distance shrinks, so selection should maximise baseline. They also disqualify candidates under *motion degeneracy* (rotation about the camera centre, leaving epipolar geometry undefined) and *structure degeneracy* (coplanar points, leaving F undetermined). Koschel et al. [5] note independently that this discards frames carrying real information and presumes a recording strategy without camera rotation about its axis. That the excluded regime is ours is not incidental: Rashidi et al. [9] construct validation sequences by deliberately filming planar walls and by rotating the camera without translation, precisely so that their selector can be shown to remove such frames.

Quality-based criteria. Koschel et al. [5] train a random forest on inertial data to reject motion-blurred and redundant frames, validated by reprojection RMSE under Meshroom. Zhang et al. [14] apply the Crete blur metric [2] to drone footage. Rashidi et al. [9] add blurring restrictions to prior selection methods. In each case blur rejection is validated against a geometric proxy, never at fixed cardinality, and never against a coverage-matched control. Koschel et al. come closest to posing the question: their Figure 1 contrasts a sequence with good image quality but no change in content against one with viewpoint variation but motion blur. They resolve it toward quality and leave overlap-based selection to future work.

Coverage-based selection. Coverage objectives are established prior art and we claim no novelty for the selection rule. Wang et al. [13] optimise a global viewing angle coverage objective at fixed cardinality via transductive experimental design; the angular bin selector we use is their comparison heuristic, dividing 360° into 36 bins and taking the best frame per bin. Zhang et al. [14] enforce a pairwise overlap rate derived from altitude and field of view. Neither work puts coverage and quality in competition: Wang et al. consider no image quality at all, and Zhang et al. fix geometric spacing first, treating blur as a local refinement within it.

These settings also guarantee angular coverage by construction: gimbal stabilised nadir flight at constant velocity [14], a rig mounted 360° camera [5], or many contributors circling a landmark [13]. Each reports coverage aware selection performing close to uniform sampling. A single handheld interior sweep is different: our unwrapped yaw trajectories span 386° to 1083° on the eight full sweeps and are non-monotonic, so temporal uniformity decouples from angular coverage.

Feed-forward multi-view reconstruction. The feed-forward family descends from DUST3R [15], which regresses pointmaps from image pairs, through unified transformers that ingest many views in a single pass [18], streaming variants with persistent state [16], and MapAnything [4], which regresses factored metric geometry from N views jointly; see [17] for a survey. Most of these anchor their output to a designated reference view, an inductive bias that Wang et al. [19] identify as a source of instability when the reference is suboptimal; their π^3 adopts a fully permutation-equivariant architecture specifically to remove that dependence and the input-ordering sensitivity it creates. Jha et al. [3] study adaptive keyframe selection for Spann3r and CUT3R, the same model family, using photometric and SSIM change signals; they assume RGB-D input, use threshold rather than fixed cardinality selection, evaluate on Chamfer distance, and include no coverage baseline. Frame selection as cardinality constrained submodular maximisation has been formalised for video language models [6]; our bin coverage objective is a coverage function and inherits the standard $(1 - 1/e)$ greedy guarantee [7], which we note for rigour rather than as a contribution.

Uncertainty in learned reconstruction. Learned depth predictors are known to be overconfident in difficult conditions [21], a problem that motivates evidential and post-hoc uncertainty methods, and post-hoc uncertainty estimation for monocular depth is an active area [20]. Uncertainty in learned priors has also been treated directly: MP-SfM [22] augments classical structure-from-motion with monocular depth and normal priors and propagates their uncertainty, reporting robustness to errors in those priors across models. Most relevant here, UfM [23] argues that single image uncertainty misses disagreement between predictions across views of the same region, and proposes multi-view disagreement as a more informative signal. Our results qualify that proposal rather than contradict it. Multi-view agreement is informative about many error sources, but it is anti-informative about the one introduced by frame selection, because a selection that loses angular coverage is also a selection whose views are mutually redundant. We are not aware of prior work reporting that a multi-view consistency score increases with reconstruction error.

3 Method

3.1 Data and features

Nine handheld phone videos of single rooms in residential and office interiors, 30 fps, one to two minutes each. From each we extract a candidate pool of 200 frames at even temporal spacing and compute two per-frame features.

Sharpness is the variance of the Laplacian, percentile normalised within a capture.

Yaw is integrated from optical flow using a nominal horizontal field of view. Since this is the input to every coverage-based strategy we validate it against camera orientations recovered from a dense reconstruction: eight of nine captures give Pearson $r \geq 0.70$ (median 0.87, four captures at $r \geq 0.98$). Optical flow systematically overestimates total rotation because translation contaminates the flow field, so we rely on the correlation rather than the absolute scale. One capture fails the check at $r = 0.154$ and we treat it separately throughout (§6).

capture	measured (m)	dense reference (m)	relative error
bedroom	3.39×3.39	3.94×3.63	11.8%
work_office	3.90×2.62	4.22×2.83	8.1%
home_office	4.72×3.72	4.04×3.41	11.4%

Table 1: Hand measured dimensions against the dense 200-frame reference, for the three captures we could physically access. Mean absolute error is 10% with no consistent direction: the reference overestimates both dimensions of bedroom and work_office and underestimates both dimensions of home_office. Room shape is nonetheless recovered, with the square room reconstructed as square. Note also that these three captures have the best self-consistency in the set (0.3 to 1.1%, Section 3.3) while showing 8 to 12% accuracy error, so internal agreement between dense reconstructions does not predict agreement with reality.

3.2 Reconstruction and metric extraction

Reconstruction uses MapAnything with mixed precision inference and edge masking. Dimensions are extracted by pooling masked world points, recovering gravity as the mean camera up-vector, aligning yaw by minimum area rectangle, and taking 5 to 95 percentile extents along the two horizontal axes.

Two checks on this procedure. The gravity estimate agrees with an independent scene layeredness estimate to 6.9° , and the resulting ceiling heights (2.24 to 2.47 m) fall in the expected range for residential interiors. The pipeline is deterministic: full reruns reproduce every reported error and failure rate to three decimals. We use the Apache-2.0 licensed `map-anything-apache` checkpoint. All reconstruction ran on a single NVIDIA GB10 with unified memory; one reconstruction takes about 2 s at $K=16$ and 5 s at $K=32$, and the full set of experiments reported here amounts to roughly 4 GPU-hours.

3.3 Reference and its noise floor

We have no tape measure ground truth. Our reference is a dense reconstruction over all 200 pool frames, so reported errors measure disagreement with a dense reconstruction rather than metric accuracy. To bound how much of that disagreement is reference noise, we reconstruct from two disjoint halves of each pool (even and odd indices, 100 frames each, both densely covering the room) and measure how much the two dense estimates differ.

Mean disagreement is 3.3% across nine captures, and 1.6% excluding the capture that also fails yaw validation, with a range of 0.3 to 5.2%. This bounds the reference’s *stability*, not its accuracy: two dense reconstructions agree closely with each other while both being wrong in the same way. We therefore hand measured the three rooms we could physically access. The dense reference differs from the tape by 8.1%, 11.4% and 11.8%, a mean absolute error of 10% with no consistent direction and roughly an order of magnitude larger than the stability figure. These same three captures have the best self-consistency in the set, so internal agreement between dense reconstructions does not predict agreement with reality. Room shape is nonetheless recovered: the square room reconstructs as square. Table 1 gives the per-room figures.

That figure sets what the reported errors can and cannot support. Our best-performing cells sit at 11.0% and 15.4% error against the reference, which is the same magnitude as the reference’s own error, so we make no claim that those reconstructions are accurate in absolute terms. The clustered cells sit at 32.4% and 43.5%, two to three times the reference uncertainty, and the spread main effect of 0.247 (95% CI [0.155, 0.340]) lies well outside it. The comparison between selection strategies is therefore supported; the absolute accuracy of any single reconstruction is not.

3.4 Selection strategies

All strategies map a feature table and budget K to K indices. **Uniform** takes even temporal spacing, the deployed baseline. **Random** samples without replacement. **Sharpness** takes the top- K by Laplacian variance. **Coverage** performs greedy marginal gain selection over 36 yaw bins of 10° , following [13]; **Coverage-FPS** substitutes farthest point sampling in yaw; **Coverage- q** restricts the greedy pool to frames above the q -th sharpness percentile.

	Relative error		Degenerate	
	high quality	low quality	high quality	low quality
spread	0.110 $\pm$ 0.077	0.154 $\pm$ 0.131	0.000	0.025
clustered	0.324 $\pm$ 0.152	0.435 $\pm$ 0.166	0.086	0.395

Table 2: 2×2 factorial at $K=16$, marginalised over view ordering. Means and standard deviations over 81 reconstructions per cell from nine captures; inference treats the capture as the unit ($n=9$).

The four factorial cells cross *spread* (greedy bin coverage) against *clustered* (a contiguous run), and *high* against *low* quality (above or below median sharpness). The informative cell is spread-low: blurry frames that nevertheless see the whole room.

Effects and experimental unit. Main effects are capture level differences of marginal means, signed so that positive values mean clustering or low sharpness increases error; the interaction is the difference of differences. Each cell contains 81 reconstructions, but these are nine captures $\times$ three seeds $\times$ three view orderings, and seeds and orderings are repeated measures within a capture. All inferential statistics therefore collapse to per-capture cell means and test across the nine captures. Cell means and standard deviations are descriptive summaries over all reconstructions; p -values and confidence intervals use $n=9$.

3.5 Metric

We report two quantities and never collapse them. **Degeneracy rate** is the fraction of reconstructions failing a geometric test fixed before any reconstruction was run: a minor dimension below 1.2 m or an aspect ratio above 4:1, neither of which describes a habitable room. **Relative dimension error** is the mean absolute relative error on the two principal horizontal extents, computed over *all* reconstructions including degenerate ones. Conditioning on success discards the worst runs; in an earlier version of this analysis it understated the worst cell’s penalty by 8 points.

3.6 View ordering

Feed-forward multi-view models anchor their output to a designated reference view and are therefore sensitive to input ordering, a dependence π^3 [19] was designed to remove. Reordering the identical sixteen frames changes the recovered long dimension by 1.5 to 13.7% per capture on the validated captures (median $\approx 9\%$), yet the between-strategy error spread exceeds the ordering variance in all nine captures. We fix temporal order for the strategy comparison and additionally marginalise the factorial over three random orderings per selection, so this variance sits inside the reported error bars.

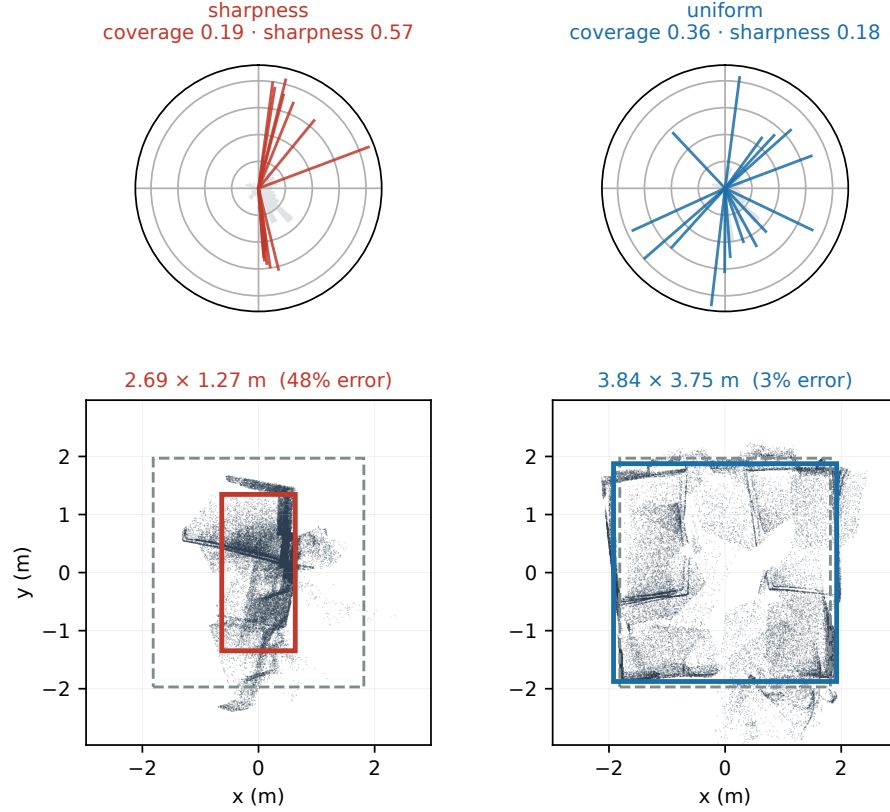
4 Results

4.1 Sharpness ranking is the worst strategy tested

At $K = 16$ across nine captures and three seeds, sharpness ranked selection gives the highest error of the seven named strategies (0.278), against 0.063 for uniform sampling and 0.114 for random, reaching 48% on the worst capture (Fig. 1). That selection achieves 0.194 angular coverage with a largest unobserved wedge of 3.35 rad, so 192° of the room is never seen. The deficit is not a small budget artefact: even at $K = 48$, sharpness reaches only 0.562 angular coverage where random sampling reaches 0.704. Ranking by sharpness is, in this capture regime, ranking by angular redundancy. This comparison fixes temporal view order rather than marginalising over it, so the large gaps hold but fine distinctions among the well-performing strategies sit within the ordering variance of §3.6 and we draw no conclusion from them.

4.2 The factorial: spread dominates, and the factors interact

Table 2 gives the result; Figure 2 in Figure 2 visualises the same contrast on a single capture. Paired across the nine captures, the spread main effect is $+0.247$ (95% CI $[+0.155, +0.340]$, $p = 0.0003$,



Dashed grey: dense reference over all 200 frames (3.94×3.63 m), registered to a common centre. bedroom, $K=16$.

Figure 1: Sharpness ranked selection against uniform sampling on one capture ($K=16$). Top: viewing directions of selected frames (colour) over the yaw distribution of all 200 candidates (grey). Bottom: reconstructed point cloud from above, with the fitted rectangle (solid) and the dense 200-frame reference (dashed), registered to a common centre. Rectangles use 5 to 95 percentile extents, so sparse peripheral points fall outside them.

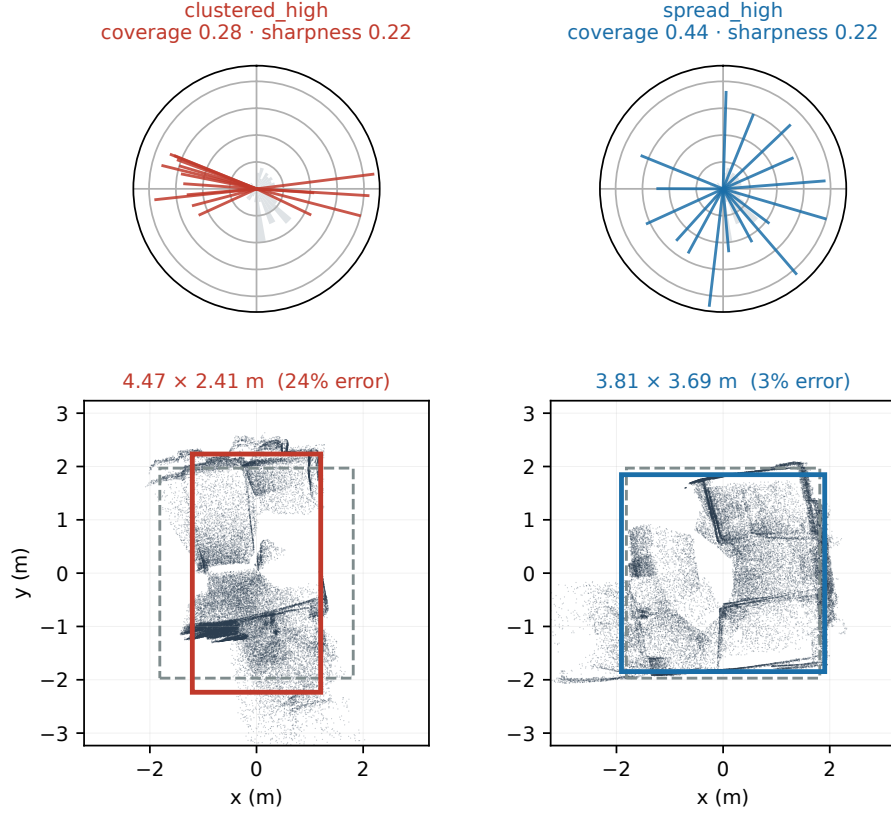
positive in 9/9) and the quality main effect $+0.077$ (95% CI $[+0.018, +0.136]$, $p = 0.017$, 8/9). Two things beyond those numbers.

First, spread-low beats clustered-high by a factor of two, in 8 of 9 captures (paired difference $+0.170$, paired t -test $p = 0.009$; Wilcoxon signed rank $p = 0.012$). Because spread and quality are manipulated independently, this directly compares reconstruction’s sensitivity to the two factors under a fixed budget: if per-frame quality were the binding constraint the ordering would reverse. The single exception is the capture that fails yaw validation (§6); on the eight validated captures the result is unanimous.

One caveat on the ratio: our spread manipulation is extreme while quality is a median split, so the factor of three reflects the contrast widths we chose rather than an intrinsic property of the factors. The direct cell comparison above does not depend on it.

Second, the cell means suggest the two factors interact. Moving from high to low quality costs 0.044 error under spread and 0.111 under clustering, and takes degeneracy from 0% to 2.5% versus 8.6% to 39.5%. The direction is consistent in 8 of 9 captures, but the interaction term is not statistically supported at the capture level ($+0.067$, 95% CI $[-0.064, +0.199]$, $p = 0.27$). We report it as a suggestive pattern that a larger capture set would be needed to establish.

Degenerate reconstructions are wrong in a different way rather than noisy: typical failures return rooms 0.59 to 0.89 m wide with unobserved wedges of 4.2 to 5.7 rad, a slab rather than a room.



Dashed grey: dense reference over all 200 frames (3.94×3.63 m), registered to a common centre. bedroom, $K=16$.

Figure 2: The factorial contrast on one capture: clustered-high against spread-high, identical mean sharpness (0.22), differing only in spread. Same conventions as Figure 1.

4.3 Budget sweep: the mechanism predicts its own boundary

If the mechanism is coverage, the advantage should shrink once coverage stops being scarce. With 36 bins, full bin coverage requires at least 36 selected frames. Table 3 sweeps K over four full sweep captures and shows two independently measured quantities collapsing at that point: the spread advantage declines gradually from 0.262 to 0.190 through $K=32$ and then halves to 0.105 at $K=48$, where our spread selections first reach coverage 1.000, and the degeneracy rate of the worst cell falls monotonically from 50% to 0% on the same schedule. Spread error is flat over the last two budgets (0.078, 0.087), so the closing gap comes from clustered selections improving. This also locates the finding, since the deployed system runs at $K=16$, near the maximum of the effect.

K	8	16	32	48
clustered error	0.421	0.316	0.268	0.192
spread error	0.160	0.104	0.078	0.087
gap	0.262	0.212	0.190	0.105
coverage achieved (spread)	0.222	0.444	0.889	1.000
degeneracy, clustered-low	0.500	0.417	0.167	0.000

Table 3: Budget sweep over the four full sweep captures available when it was run, three seeds each, so twelve reconstructions per cell. $K=12$ and $K=24$ are omitted for space and interpolate smoothly.

	relative error	multi-view confidence
spread, high quality	0.110	0.444
spread, low quality	0.154	0.363
clustered, high quality	0.324	0.669
clustered, low quality	0.435	0.674

Table 4: Multi-view consistency score against error, means over 81 reconstructions per cell. The ordering is inverted throughout.

4.4 Confidence increases with error

Cell means are given in Table 4. The signal available at inference time is a multi-view consistency score: the agreement between per-view depth predictions where they overlap. Practitioners naturally read it as a reliability estimate. It *increases* with error. Within each capture, the Spearman correlation between confidence and error is positive in 9 of 9 captures, mean $\rho = +0.529$, individually significant in 8 (sign test across captures $p = 0.004$; Wilcoxon $p = 0.004$). Pooling reconstructions across captures weakens the apparent relationship rather than strengthening it, because between-capture difficulty varies more than confidence does. Degenerate reconstructions carry a 41% higher mean consistency score than valid ones from the same capture (0.710 vs 0.505, paired across the eight captures containing both, $p = 0.003$, 7/8 captures).

This is not a defect in the score so much as a mismatch between what it measures and what a practitioner would use it for. Clustered views observe the same surfaces from similar angles, so their depth predictions agree, and agreement is exactly what the score reports. Consistency is not correctness. Under frame selection the two move in opposite directions, because the selections that lose coverage are the same selections that gain redundancy.

For a deployed system the implication is uncomfortable. A pipeline that selects frames by sharpness, a defensible looking choice with substantial literature behind it, will produce systematically wrong geometry while reporting elevated confidence.

5 Is an image-only VLM affected by the same selection?

Geometric reconstruction sits upstream of the spatial reasoning vision-language models perform on indoor scenes. We expected selection errors to propagate. In our room dimension task they do not, and the tested VLM offers no compensating reliability signal either.

We gave a vision-language model (claude-sonnet-5, queried August 2026) the same selected frames and asked it to estimate the room’s floor dimensions and its own confidence, over all nine captures, seven strategies and three seeds (189 queries).

Frame selection barely affects the VLM. Strategy means span 0.184 to 0.270 with standard deviations of 0.07 to 0.17, so the between-strategy spread is smaller than the within-strategy noise. The 2×2 ordering is even mildly inverted relative to the geometric result (clustered 0.213, spread 0.248), and sharpness-ranked selection, the worst strategy for reconstruction, gives the lowest VLM error of the seven.

VLM error is capture dominated. Per-capture mean error ranges from 0.122 to 0.541, while three captures have within-capture standard deviations of 0.018, 0.031 and 0.036 across all seven strategies. On one capture the VLM returns essentially the same estimate whatever frames it is shown (0.188 ± 0.018). Which room is being estimated explains substantially more variance than which frames are selected. Geometric and VLM error are also uncorrelated across the 63 matched capture-strategy cells (Spearman -0.127 , $p = 0.32$; Figure 3a).

One possible explanation is that the two systems solve the task by different routes. MapAnything aggregates geometry across views and therefore needs the views to cover the room, whereas a VLM may rely more heavily on visual priors such as furniture scale, door and ceiling heights, and

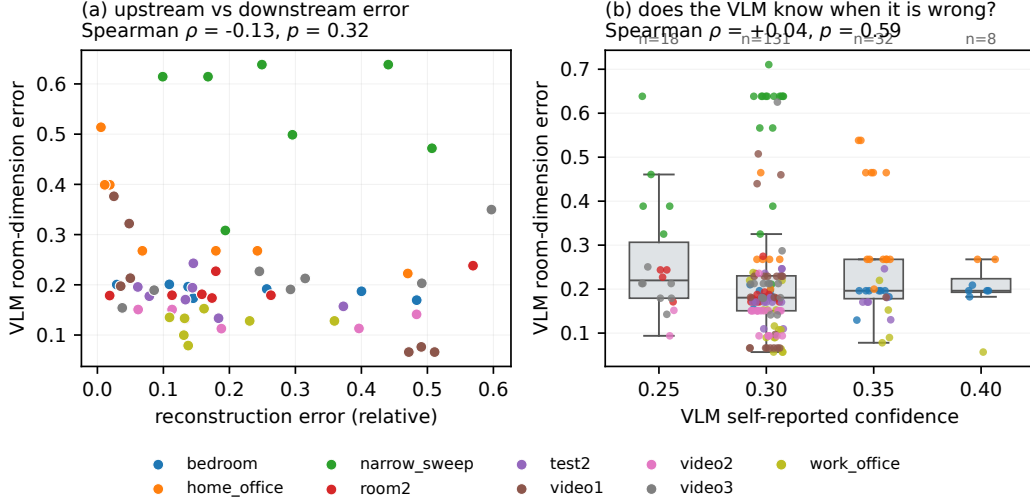


Figure 3: Neither downstream signal tracks the upstream failure. (a) Per capture and strategy, reconstruction error against the VLM’s room-dimension error: the selections that break the geometry are not the ones the VLM gets wrong. (b) Error distribution at each of the four confidence values the model actually reports, with individual queries jittered over the boxes. Of 189 queries, 131 return a confidence of exactly 0.30 and the 0.40 bin contains only eight, so the outer bins should not be over-read. Colour denotes capture. Section A gives the query text.

perspective, which one adequate view can supply. We did not measure what the model attends to, so this remains a hypothesis.

VLM confidence provides no reliable error signal. Across all 189 queries the model emits only four distinct confidence values, 0.25, 0.30, 0.35 and 0.40, so per-strategy means sit in a narrow band from 0.287 to 0.320 while error ranges from 0.184 to 0.270. Confidence and error are uncorrelated (Spearman $+0.039$, $p = 0.59$), and the error distribution is essentially unchanged across the four reported levels (Figure 3b). The signal is not merely miscalibrated; it barely varies. The prompt asked for a checkable quantity, the probability that the estimate falls within 15% of the true value (Appendix A), so this is not a failure to interpret a vague question.

We note also that a pooled correlation between angular coverage and VLM error that appeared strong on a five-capture subset ($\rho = -0.37$) falls to $\rho = -0.125$ ($p = 0.09$) on the full set, consistent with it having been between-capture variance rather than a selection effect.

Taken with §4.4, the deployment picture is that neither signal in the stack was usable in our experiments. The reconstruction model’s consistency score is highest where accuracy is lowest. The VLM’s stated confidence carries no usable information about its own error, and its errors are unrelated to the geometry’s, so in this task it could not act as a check on a corrupted reconstruction either.

6 Limitations

No absolute ground truth. Errors measure disagreement with a dense reconstruction from the same model, not metric accuracy; Section 3.3 bounds its noise at 3.3% and its accuracy at 10%. Two findings do not depend on the reference: degeneracy rates rest on physically impossible dimensions, and the confidence results are internal to the models. Uniform sampling is also structurally advantaged as the K -frame subsample most resembling the reference, so we make no claim about it being optimal.

Nine single room captures, three with physical ground truth. All are residential or office interiors. The remaining six were not accessible for measurement. The effect is consistent across them, but the sample is small and we do not test unusual room shapes.

One capture fails yaw validation and is anomalous throughout. A narrow sweep capture (80° unwrapped) gives $r = 0.154$ against reconstructed camera orientations, 16.4% reference disagreement

against a 1.6% mean elsewhere, 0.541 VLM error against 0.122 to 0.334 elsewhere, the largest ordering variance (18.7%), and is the only capture in our set where spread-low loses to clustered-high. We report it in aggregates but draw no boundary condition from it. Characterising behaviour under narrow sweeps needs captures whose yaw estimate validates.

Scope of the experiments. The strategy comparison, factorial, consistency score, reference, yaw and downstream VLM analyses use all nine captures; the budget sweep uses five. We also use a single sharpness measure, Laplacian variance, whereas the works we contrast with use a perceptual blur metric [2] or DFT magnitude [5]; the factorial partitions at the median and is largely insensitive to this, but the top- K sharpness strategy could rank frames differently under a directional measure.

Implementation gap with the deployed system. Our experiments use the PyTorch reference implementation, because the deployed ONNX export is fixed at 16 views and pads shorter inputs by repeating the last frame, which would confound any comparison at $K \neq 16$. On one capture the deployed service returns 3.63×3.60 m where our dense reference returns 3.94×3.63 m, about 8% apart. All our claims are relative comparisons within one implementation, so this does not affect them, but our absolute values should not be read as the deployed system’s.

Propagation through the reconstruction is untested. Section 5 gives the VLM the same frames, not the reconstruction, so it tests whether an image-only VLM shares the selection sensitivity, not whether corrupted geometry poisons a system reasoning over it.

One reconstruction model. Whether the finding transfers to other feed-forward multi-view backbones, particularly permutation-equivariant ones, is open. It would also test whether the order sensitivity we measure is intrinsic to the architecture class.

7 Conclusion

Frame selection for feed-forward multi-view reconstruction is not the same problem as keyframe selection for structure-from-motion, and the inherited heuristic of keeping the sharp frames is harmful when the budget is small. The cause is not subtle: sharp frames cluster where the camera moved slowly, and a room you did not look at cannot be reconstructed. Under a fixed budget, the estimated main effect of viewpoint coverage is roughly three times that of per-frame image quality, and blurry frames that see the whole room beat sharp frames that see one wall in 8 of 9 captures.

For deployment the reliability consequence matters more than the accuracy one. An upstream heuristic with substantial literature behind it can silently corrupt the geometry a multimodal system reasons over, and neither uncertainty signal available at inference gives warning: the reconstruction model’s consistency score is highest exactly when it is most wrong, and in our experiment a VLM’s spatial estimates were insensitive to the selection that broke the geometry, with confidence that carried no error signal. In a pipeline where semantic reasoning is grounded in reconstructed geometry, that is a failure mode with no detector. The practical guidance follows: when the budget is small relative to the angular extent of the scene, select for coverage and let sharpness follow, and do not treat either confidence signal as a proxy for correctness.

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A VLM query details

The downstream experiment in Section 5 queries `claude-sonnet-5` through its public API. Each query contains the $K=16$ selected frames in temporal order, resized so the long side is 768 px and re-encoded as JPEG at quality 85, followed by the text below. Output is capped at 300 tokens. Responses are parsed as JSON; a response that fails to parse is retried up to four times with exponential backoff and recorded as a failure if it never parses.

```
These are 16 frames sampled from a single continuous video of ONE
room, in temporal order.

Estimate the room's floor dimensions.

Reply with ONLY a JSON object, no other text:
{"length_m": <number>, "width_m": <number>, "ceiling_m": <number>,
"confidence": <0-1>, "reasoning": "<one sentence>"}

length_m is the longer horizontal dimension, width_m the shorter,
both in metres. confidence is how sure you are that your estimate
is within 15% of the true value.
```

We report `length_m` and `width_m` sorted so the larger is first, matching the convention used for the geometric metric, and `confidence` as the self-reported reliability estimate. The `ceiling_m` and `reasoning` fields were recorded but are not analysed here.

Note that the confidence field asks for a specific, checkable quantity, namely the probability that the estimate falls within 15% of the true value, rather than a vague self-assessment. The reported values cluster between 0.29 and 0.33 across every condition, which is part of why they carry so little information about the actual error.